GR 313

1) Natura seriei $\sum_{n\geq 1} \frac{2\cdot 5\cdot 8\cdot \ldots\cdot (3n-1)}{1\cdot 5\cdot 9\cdot \ldots\cdot (4n-3)}(\alpha-3)^n$, $\alpha>3$

$x_n = \frac{2\cdot 5\cdot \ldots\cdot (3n-1)}{1\cdot 5\cdot \ldots\cdot (4n-3)}$ Conform crit rap $\Rightarrow$

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n} = \lim_{n\to\infty}\frac{2\cdot 5\cdot \ldots\cdot (3n-1)(3n+2)}{1\cdot 5\cdot \ldots\cdot (4n-3)(4n+1)}\cdot\frac{1\cdot 5\cdot \ldots\cdot (4n-3)}{2\cdot 5\cdot \ldots\cdot (3n-1)}\cdot\frac{(\alpha-3)^{n+1}}{(\alpha-3)^n}$$

$$= \lim_{n\to\infty}\frac{3n+2}{4n+1}\cdot(\alpha-3) = \frac{3}{4}\cdot(\alpha-3)$$

I $\frac{3}{4}(\alpha-3)<1 \Rightarrow \alpha-3<\frac{4}{3} \Rightarrow \alpha<\frac{4}{3}+3 \Rightarrow \alpha<\frac{13}{3}$

$\sum_{n\geq 1} x_n$ convergentă

II $\frac{3}{4}(\alpha-3)>1 \Rightarrow \alpha>\frac{13}{3} \Rightarrow \sum_{n\geq 1} x_n$ divergentă $\alpha=\frac{4}{3}+3$

III $\frac{3}{4}(\alpha-3)=1 \Rightarrow \alpha=\frac{13}{3} \Rightarrow \sum_{n\geq 1} x_n = \sum_{n\geq 1}\frac{2\cdot 5\cdot \ldots\cdot (3n-1)}{1\cdot 5\cdot \ldots\cdot (4n-3)}\cdot\left(\frac{4}{3}\right)^n$

Crit. R-D. $\Rightarrow \lim_{n\to\infty} n\left(\frac{x_n}{x_{n+1}}-1\right) = \lim_{n\to\infty} n\cdot\left(\frac{4n+1}{(3n+2)(\alpha-3)}-1\right)$

$$= \lim_{n\to\infty} n\cdot\left[\frac{4n+1}{(3n+2)\frac{4}{3}}-1\right] = \lim_{n\to\infty} n\,\frac{3(4n+1)-4(3n+2)}{4(3n+2)}$$

$$= \lim_{n\to\infty} n\cdot\frac{12n+3-12n-8}{12n+8} = \lim_{n\to\infty}\frac{-5n}{12n+8} = -\frac{5}{12}<1$$

$\Rightarrow \sum_{n\geq 1} x_n$ divergentă pt. $\alpha=\frac{13}{3}$

2) Convergența și absolut convergența seriei

$$\sum_{n\geq 2} \frac{(-1)^{n-1}}{\sqrt[3]{n+1}\,\ln^2(n+2)} = \sum_{n\geq 2} x_n$$

AC

$$\sum_{n\geq 2} |x_n| = \sum_{n\geq 2} \frac{1}{\sqrt[3]{n+1}\,\ln^2(n+2)}$$

Crit. log $\Rightarrow \lim_{n\to\infty} \frac{\ln\left(\frac{1}{x_n}\right)}{\ln n} = \lim_{n\to\infty} \frac{\ln\left[\sqrt[3]{n+1}\,\ln^2(n+2)\right]}{\ln n}$

$$= \lim_{n\to\infty} \frac{\ln\left(\sqrt[3]{n+1}\right) + \ln\left(\ln^2(n+2)\right)}{\ln n} =$$

$$= \lim_{n\to\infty} \frac{\frac{1}{3}\ln(n+1) + \ln\left(2\ln(n+2)\right)}{\ln n} = \frac{1}{3} < 1$$

$\frac{1}{3}$

0

$\Rightarrow \sum_{n\geq 2} |x_n|$ nu e absolut convergentă

C

$$\alpha_n = \frac{1}{\sqrt[3]{n+1}\,\ln^2(n+2)}$$

$\lim_{n\to\infty} \alpha_n = 0$

$\alpha_n > 0$

$\Rightarrow$ Crit Leib.

$$\frac{\alpha_{n+1}}{\alpha_n} = \frac{\sqrt[3]{n+1}\,\ln^2(n+2)}{\sqrt[3]{n+2}\,\ln^2(n+3)} < 1 \Rightarrow \alpha_n \searrow$$

$\Rightarrow \sum_{n\geq 1} x_n = \sum_{n\geq 1} (-1)^{n-1} \alpha_n$ e convergentă

3) Suma seriei $\sum_{n\geq 2} \frac{n+1}{3^n \cdot n!}$ cu ajutorul seriilor de puteri

$$e^x = \sum_{n\geq 0} \frac{1}{n!} \cdot x^n$$

$$\sum_{n\geq 2} \frac{n+1}{3^n \cdot n!} = \sum_{n\geq 2} \frac{n}{3^n \cdot n!} + \sum_{n\geq 2} \frac{1}{3^n \cdot n!}$$

$$\sum_{n\geq 2} \frac{n}{3^n \cdot n!} = \sum_{n\geq 2} \frac{1}{3^n (n-1)!} = \sum_{n\geq 1} \frac{1}{3^{n+1} \cdot n!} = \frac{1}{3} \sum_{n\geq 1} \frac{1}{3^n \cdot n!}$$

$$e^{1/3} = \sum_{n\geq 0} \frac{1}{3^n \cdot n!} = \frac{1}{3^0 \cdot 0!} + \sum_{n\geq 1} \frac{1}{3^n \cdot n!} \Rightarrow \boxed{\sum_{n\geq 1} \frac{1}{3^n \cdot n!} = e^{1/3} - 1}$$

$$e^{1/3} = \sum_{n\geq 0} \frac{1}{3^n \cdot n!} = 1 + \frac{1}{3 \cdot 1!} + \sum_{n\geq 2} \frac{1}{3^n \cdot n!}$$

$$\Rightarrow \boxed{\sum_{n\geq 2} \frac{1}{3^n \cdot n!} = e^{1/3} - \frac{4}{3}}$$

$$\sum_{n\geq 2} \frac{n+1}{3^n \cdot n!} = \frac{1}{3} \cdot \left(e^{1/3} - 1\right) + e^{1/3} - \frac{4}{3} = \frac{4}{3} e^{1/3} - \frac{5}{3} = \frac{4e^{1/3} - 5}{3}$$

4) Aflați derivatele parțiale pt funcția

$$f(x,y) = \frac{y^5 - x^2 y^3}{(x^2+y^2)^2}$$

$$\frac{\partial f}{\partial x} = \frac{-2xy^3 (x^2+y^2)^2 - (y^5 - x^2 y^3) 2 (x^2+y^2) \cdot 2x}{(x^2+y^2)^4}$$

$$\frac{\partial f}{\partial x} = \frac{-2xy^3(x^2+y^2) - (y^5 - x^2y^3)\cdot 4x}{(x^2+y^2)^3} = \frac{-2x\left[y^3x^2 + y^5 + 2y^5 - 2x^2y^3\right]}{(x^2+y^2)^3}$$

$$\frac{\partial f}{\partial x} = \frac{-2x\left[3y^5 - x^2y^3\right]}{(x^2+y^2)^3} = \frac{-2xy^3(3y^2 - x^2)}{(x^2+y^2)^3}$$

$$\frac{\partial f}{\partial y} = \frac{(5y^4 - 3x^2y^2)(x^2+y^2)^2 - (y^5 - x^2y^3)2(x^2+y^2)\cdot 2y}{(x^2+y^2)^4}$$

$$\frac{\partial f}{\partial y} = \frac{y^2(5y^2 - 3x^2)(x^2+y^2) - 4y^4(y^2 - x^2)}{(x^2+y^2)^3} =$$

$$\frac{\partial f}{\partial y} = \frac{y^2\left[5x^2y^2 + 5y^4 - 3x^4 - 3x^2y^2\right] - 4y^2(y^4 - x^2y^2)}{(x^2+y^2)^3}$$

$$\frac{\partial f}{\partial y} = \frac{y^2\left[5x^2y^2 + 5y^4 - 3x^4 - 3x^2y^2 - 4y^4 + 4x^2y^2\right]}{(x^2+y^2)^3}$$

$$\frac{\partial f}{\partial y} = \frac{y^2\left[6x^2y^2 + y^4 - 3x^4\right]}{(x^2+y^2)^3}$$

5) Determinati extremele locale pt. $f: \mathbb{R}^2 \to \mathbb{R}$

$$f(x,y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$$

$$\begin{cases} \frac{\partial f}{\partial x} = 4x^3 - 4x + 4y = 0 \\ \frac{\partial f}{\partial y} = 4y^3 + 4x - 4y = 0 \end{cases} \Rightarrow (+)\ 4x^3 + 4y^3 = 0 \Rightarrow x^3 + y^3 = 0$$

$$(x+y)(x^2 - xy + y^2) = 0$$

I $y=-x \Rightarrow 4x^3-4x-4x=0 \Rightarrow 4x^3-8x=0 \Rightarrow$

$\Rightarrow 4x(x^2-2)=0 \Rightarrow x_1=0 \Rightarrow y_1=0 \Rightarrow M_1(0;0)$

$x_2=\sqrt{2} \Rightarrow y_2=-\sqrt{2} \Rightarrow M_2(\sqrt{2};-\sqrt{2})$

$x_3=-\sqrt{2} \Rightarrow y_3=\sqrt{2} \Rightarrow M_3(-\sqrt{2};\sqrt{2})$

II $x^2-xy+y^2=0 \Rightarrow \begin{cases} x^2-xy+y^2=0 \\ y=x-x^3=x(1-x^2) \end{cases} \Rightarrow$

$\Rightarrow x^2-x^2(1-x^2)+x^2(1-x^2)^2=0 \Rightarrow x^2[1-1+x^2+1-2x^2+x^4]=0$

$\Rightarrow x^2[x^4-x^2+1]=0$

(a) $\Delta=1-4\cdot1\cdot1<0 \Rightarrow$ nu are sol. reale.

(b) $x^2=0 \Rightarrow x=0 \Rightarrow y=0 \Rightarrow M_1(0;0)$

$\frac{\partial^2 f}{\partial x^2}=12x^2-4 \qquad \frac{\partial^2 f}{\partial y^2}=12y^2-4 \qquad \frac{\partial^2 f}{\partial x \partial y}=4$

1 $H(0,0)=\begin{pmatrix} -4 & 4 \\ 4 & -4 \end{pmatrix} \quad \Delta_1<0 \quad \Delta_2=0$

$\det(H-\lambda J)=0 \Rightarrow \begin{vmatrix} -4-\lambda & 4 \\ 4 & -4-\lambda \end{vmatrix}=0 \Rightarrow (4+\lambda)^2-16=0$

$\Rightarrow 16+8\lambda+\lambda^2-16=0 \Rightarrow \lambda(\lambda+8)=0$

$f(x,y)-f(0,0)=\frac{1}{1!}[0\cdot(x-0)+0\cdot(y-0)]+\frac{1}{2!}[-4(x-0)^2$

$+(-4)(y-0)^2+2\cdot4(x-0)(y-0)]$

$$f(x,y)-f(0,0)=\frac{1}{2!}\left[-4x^2-4y^2+8xy\right]=\frac{-4}{2}\left[x^2+y^2-2xy\right]$$

$$=-2(x-y)^2<0 \Rightarrow \underline{M_1(0;0) \text{ pct. de maxim}}$$

2

3 $$H(\pm\sqrt{2};\mp\sqrt{2})=\begin{pmatrix}20 & 4\\ 4 & 20\end{pmatrix} \qquad \begin{matrix}\Delta_1>0\\ \Delta_2>0\end{matrix} \Rightarrow$$

$\Rightarrow \underline{M_2(\sqrt{2};-\sqrt{2}) \text{ pct. de minim}}$

$\underline{M_3(-\sqrt{2};+\sqrt{2}) \text{ pct. de minim.}}$

GR. 314

1) Natura seriei $\sum_{n\geq 1} \frac{1\cdot 6\cdot 11\cdot \ldots\cdot (5n-4)}{1\cdot 7\cdot 13\cdot \ldots\cdot (6n-5)} (\alpha+1)^n$, $\alpha > -1$

$$\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = \lim_{n\to\infty} \frac{1\cdot 6\cdot \ldots\cdot (5n-4)(5n+1)}{1\cdot 7\cdot \ldots\cdot (6n-5)(6n+1)} \cdot \frac{1\cdot 7\cdot \ldots\cdot (6n-5)}{1\cdot 6\cdot \ldots\cdot (5n-4)} \cdot \frac{(\alpha+1)^{n+1}}{(\alpha+1)^n}$$

$$= \lim_{n\to\infty} \frac{5n+1}{6n+1}\cdot(\alpha+1) = \frac{5}{6}(\alpha+1)$$

Criteriul Raportului $\Rightarrow$

I $\frac{5}{6}(\alpha+1) > 1 \Rightarrow \alpha+1 > \frac{6}{5} \Rightarrow \alpha > \frac{1}{5} \Rightarrow \sum_{n\geq 1} x_n$ diverg.

II $\frac{5}{6}(\alpha+1) < 1 \Rightarrow \alpha < \frac{1}{5} \Rightarrow \sum_{n\geq 1} x_n$ conv.

III $\frac{5}{6}(\alpha+1) = 1 \Rightarrow \alpha = \frac{1}{5}$

Criteriul Raabe Duhamel $\Rightarrow$

$$\Rightarrow \lim_{n\to\infty} n\left[\frac{6n+1}{(5n+1)\left(\frac{1}{5}+1\right)} - 1\right] = \lim_{n\to\infty} n \cdot \frac{6n+1-(5n+1)\cdot\frac{6}{5}}{(5n+1)\cdot\frac{6}{5}}$$

$$= \lim_{n\to\infty} n\cdot \frac{6n+1-\left(6n+\frac{6}{5}\right)}{6n+\frac{6}{5}} = \lim_{n\to\infty} n\cdot \frac{6n+1-6n-\frac{6}{5}}{6n+\frac{6}{5}}$$

$$= \lim_{n\to\infty} \frac{-\frac{1}{5}n}{6n+\frac{6}{5}} = \frac{-\frac{1}{5}}{6} = -\frac{1}{30} < 1 \Rightarrow \sum_{n\geq 1} x_n \text{ diverg.}$$

2) Convergența și absolut convergența seriei

$$\sum_{n\geq 1} \frac{(-1)^{n-1}}{\sqrt[4]{n}\cdot \ln^3(n+1)} = \sum_{n\geq 1} x_n$$

A.C. $\sum_{n\geq 1}|x_n| = \sum_{n\geq 1}\frac{1}{\sqrt[4]{n}\ln^3(n+1)}$

Crit. logar. $\Rightarrow \lim_{n\to\infty}\frac{\ln\frac{1}{x_n}}{\ln n} = \lim_{n\to\infty}\frac{\ln\left[\sqrt[4]{n}\ln^3(n+1)\right]}{\ln n}$

$= \lim_{n\to\infty}\frac{\frac{1}{4}\ln n + \ln\left[3\ln(n+1)\right]}{\ln n} = \frac{1}{4} < 1 \Rightarrow$

$\Rightarrow \sum_{n\geq 1}|x_n|$ diverg.

$\Rightarrow \sum_{n\geq 1} x_n$ nu e A.C.

C. fie $a_n = \frac{1}{\sqrt[4]{n}\ln^3(n+1)}$ $\quad a_n \geq 0$

$\lim_{n\to\infty} a_n = 0$ $\quad$ Crit. Leibniz $\Rightarrow$

$\frac{a_{n+1}}{a_n} = \frac{\sqrt[4]{n}\ln^3(n+1)}{\sqrt[4]{n+1}\ln^3(n+2)} < 1 \Rightarrow a_n \searrow$

$\Rightarrow \sum_{n\geq 1}(-1)^{n-1}a_n$ conv. $\Rightarrow \sum_{n\geq 1} x_n$ conv.

3) Aflați suma seriei $\sum_{n\geq 1}\frac{(-1)^n\cdot 11^n\cdot n}{(n+1)!}$ cu ajutorul seriilor de puteri

$e^x = \sum_{n\geq 0}\frac{1}{n!}x^n \xrightarrow{\text{int.}} e^x = \sum_{n\geq 0}\frac{1}{n!}\frac{x^{n+1}}{n+1} = \sum_{n\geq 0}\frac{1}{(n+1)!}x^{n+1}$

$e^x = \sum_{n\geq 0}\frac{1+n-n}{(n+1)!}x^{n+1} = \sum_{n\geq 0}\frac{n+1}{(n+1)!}x^{n+1} - \sum_{n\geq 0}\frac{n}{(n+1)!}x^{n+1}$

$e^x = \sum_{n\geq 0}\frac{1}{n!}x^{n+1} - \sum_{n\geq 0}\frac{n}{(n+1)!}x^{n+1}$

$$e^x = \sum_{n\geq 0} \frac{1}{n!}\cdot x^n \cdot x - \left[0 + \sum_{n\geq 1} \frac{n}{(n+1)!} x^{n+1}\right]$$

$$e^x = e^x \cdot x - \sum_{n\geq 1} \frac{n}{(n+1)!} x^{n+1}$$

$$e^{-11} = e^{-11}\cdot(-11) - \sum_{n\geq 1} \frac{n}{(n+1)!}(-11)^{n+1} = -11e^{-11} + 11\sum_{n\geq 1} \frac{(-1)^n n}{(n+1)!} 11^n$$

$$11S(x) - 11e^{-11} = e^{-11} \Rightarrow S(x) = \frac{e^{-11}+11e^{-11}}{11} = \frac{12e^{-11}}{11}$$

$$S(x) = \frac{12}{11e^{11}}$$

4) Aflați derivatele parțiale pt. funcția

$$f(x,y) = \frac{3x^3y^2 + xy^4}{(x^2+y^2)^2}$$

$$\frac{\partial f}{\partial x} = \frac{(9y^2x^2 + y^4)(x^2+y^2)^2 - (3x^3y^2 + xy^4)2(x^2+y^2)2x}{(x^2+y^2)^4}$$

$$\frac{\partial f}{\partial x} = \frac{y^2(9x^2+y^2)(x^2+y^2) - 4x^2y^2(3x^2+y^2)}{(x^2+y^2)^3}$$

$$\frac{\partial f}{\partial x} = \frac{y^2(9x^4+9x^2y^2+x^2y^2+y^4) - y^2(12x^4+4x^2y^2)}{(x^2+y^2)^3}$$

$$\frac{\partial f}{\partial x} = \frac{y^2(9x^4+10x^2y^2+y^4-12x^4-4x^2y^2)}{(x^2+y^2)^3} = \frac{y^2(y^4+6x^2y^2-3x^4)}{(x^2+y^2)^3}$$

$$\frac{\partial f}{\partial y} = \frac{(6x^3y+4xy^3)(x^2+y^2)^2 - (3x^3y^2+xy^4)2(x^2+y^2)2y}{(x^2+y^2)^4}$$

$$\frac{\partial f}{\partial y} = \frac{2xy(3x^2+2y^2)(x^2+y^2) - 2xy\cdot 2(3x^2y^2+y^4)}{(x^2+y^2)^3}$$

$$\frac{\partial f}{\partial y} = \frac{2xy[3x^4+3x^2y^2+2x^2y^2+2y^4-6x^2y^2-2y^4]}{(x^2+y^2)^3}$$

$$\frac{\partial f}{\partial y} = \frac{2xy[3x^4-x^2y^2]}{(x^2+y^2)^3} = \frac{6x^5y-2x^3y^3}{(x^2+y^2)^3}$$

5) Determinați extremele locale pt. $f: \mathbb{R}^2 \to \mathbb{R}$

$$f(x,y) = 4xy - x^4 - y^4$$

$$\begin{cases} \frac{\partial f}{\partial x} = 4y - 4x^3 = 0 \\ \frac{\partial f}{\partial y} = 4x - 4y^3 = 0 \end{cases} \Rightarrow \begin{cases} y - x^3 = 0 \Rightarrow y = x^3 \\ x - y^3 = 0 \end{cases} \Rightarrow$$

$$\Rightarrow x - x^9 = 0 \Rightarrow x(1-x^8) = 0$$

$$\Rightarrow x(1+x^4)(1+x^2)(1+x)(1-x) = 0$$

I $x = 0 \Rightarrow y = 0 \quad M_1(0;0)$

II $1+x=0 \Rightarrow x=-1 \Rightarrow y=-1 \quad M_2(-1;-1)$

III $1-x=0 \Rightarrow x=1 \Rightarrow y=1 \quad M_3(1;1)$

$$\frac{\partial^2 f}{\partial x \partial y} = 4 \qquad \frac{\partial^2 f}{\partial x^2} = -12x^2 \qquad \frac{\partial^2 f}{\partial y^2} = -12y^2$$

1 $H(0;0) = \begin{pmatrix} 0 & 4 \\ 4 & 0 \end{pmatrix}$

$$\det(H-\lambda I)=0 \Rightarrow \begin{vmatrix} -\lambda & 4 \\ 4 & -\lambda \end{vmatrix}=0 \Rightarrow \lambda^2-16=0 \Rightarrow$$

$$\Rightarrow \lambda=\pm 4$$

$\Rightarrow$ (0;0) nu e pct. de extrem

2 $H(\pm 1;\pm 1)=\begin{pmatrix} -12 & 4 \\ 4 & -12 \end{pmatrix}$ $\quad \Delta_1<0$

$\Delta_2=144-16>0$

$\Delta_1\cdot(-1)>0$
$\Delta_2\cdot(-1)^2>0$ $\Rightarrow$ (-1;-1) pct. de maxim

(1;1) pct. de maxim

Subiect IS.

1) Aflați natura seriei $\sum_{n\geq 1} \frac{2+n^\alpha}{n-\sqrt[3]{n^3-n^2}}$

$$\sum_{n\geq 1} \frac{2+n^\alpha}{n-\sqrt[3]{n^3-n^2}} = \sum_{n\geq 1} \frac{(2+n^\alpha)\left[n^2+n\sqrt[3]{n^3-n^2}+\sqrt[3]{(n^3-n^2)^2}\right]}{n^3-n^3+n^2}$$

$$= \sum_{n\geq 1} (2+n^\alpha)\left[1+\sqrt[3]{1-\frac{1}{n}}+\sqrt[3]{\left(1-\frac{1}{n}\right)^2}\right] = \sum_{n\geq 1} x_n$$

fie $y_n = \sum_{n\geq 1} \frac{1}{n^{-\alpha}}$

$$\lim_{n\to\infty} \frac{x_n}{y_n} = \lim_{n\to\infty} (2+n^\alpha)\left[1+\sqrt[3]{1-\frac{1}{n}}+\sqrt[3]{\left(1-\frac{1}{n}\right)^2}\right]\cdot n^{-\alpha}$$

$$= \lim_{n\to\infty} \left(\frac{2}{n^\alpha}+1\right)\left[1+\sqrt[3]{1-\frac{1}{n}}+\sqrt[3]{\left(1-\frac{1}{n}\right)^2}\right] = 1\cdot 3 = 3 \in (0;+\infty)$$

$\xrightarrow[\text{III}]{\text{Crit}}$ $\sum_{n\geq 1} x_n \sim \sum_{n\geq 1} y_n$

$\sum_{n\geq 1} y_n$ serie Riemann

$\Rightarrow$ pt. $-\alpha > 1 \Rightarrow \alpha < -1$ $\sum_{n\geq 1} y_n$ converg.

$\Rightarrow$ pt. $-\alpha \leq 1 \Rightarrow \alpha \geq -1$ $\sum_{n\geq 1} y_n$ diverg.

$\Rightarrow$

$\Rightarrow$ pt. $\alpha < -1$ $\sum_{n\geq 1} x_n$ converg.

pt. $\alpha \geq -1$ $\sum_{n\geq 1} x_n$ diverg.

2) Aflați mulțimea de convergență și suma seriei

$$\sum_{n\geq 2} \frac{1}{n} x^{2n} = \sum_{n\geq 2} a_n \cdot x^{2n}$$

$$\ell = \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{1}{n+1}\cdot n = 1 \Rightarrow R = \frac{1}{\ell} = 1$$

$$\Rightarrow \mathcal{I}_c = (-1;1)$$

$$x = \pm 1 \Rightarrow \sum_{n\geq 2} \frac{1}{n}\cdot(\pm 1)^{2n} = \sum_{n\geq 2} \frac{1}{n} \text{ diverg.}$$

$$\Rightarrow M_C = (-1;1)$$

$$S(x) = \sum_{n\geq 2} \frac{1}{n} x^{2n} \Big/ \Rightarrow f(y) = \sum_{n\geq 2} \frac{1}{n}\cdot y^n \xrightarrow{\text{deriv.}} f'(y) = \sum_{n\geq 2} y^{n-1} \Big/$$

Notăm $y = x^2$

$$\Rightarrow f'(y) = \sum_{n\geq 1} y^n \Big| \Rightarrow f'(y) = \frac{1}{1-y} - 1$$

$$\sum_{n\geq 0} y^n = \frac{1}{1-y} \Rightarrow 1 + \sum_{n\geq 1} y^n = \frac{1}{1-y}$$

$$\xrightarrow{\text{integ.}} f(y) = \int_0^y \frac{1}{1-t} - 1\, dt = \left[-\ln(1-t) - t\right]\Big|_0^y \Rightarrow$$

$$f(y) = -\ln(1-y) - y \Rightarrow S(x) = -\ln(1-x^2) - x^2$$

3) Aproximați $\varepsilon < 10^{-3}$ integrala $\int_0^{1/3} \frac{1-\cos x}{x^2}\, dx$

$$\frac{1-\cos x}{x^2} = \frac{1 - \sum_{n\geq 0} \frac{(-1)^n}{(2n)!} x^{2n}}{x^2} = \frac{x - \left(x + \sum_{n\geq 1} \frac{(-1)^n}{(2n)!} x^{2n}\right)}{x^2}$$

$$= \sum_{n\geq 1} \frac{(-1)^{n+1}}{(2n)!} x^{2n-2}$$

$$\int_0^{1/3} \frac{1-\cos x}{x^2}\, dx = \int_0^{1/3} \sum_{n\geq 1} \frac{(-1)^{n+1}}{(2n)!} x^{2n-2}\, dx = \sum_{n\geq 1} \frac{(-1)^{n+1}}{(2n)!} \frac{x^{2n-1}}{2n-1}\Big|_0^{1/3}$$

$$= \sum_{n\geq 1} \frac{(-1)^{n+1}}{(2n)!} \cdot \frac{1}{3^{2n-1}(2n-1)}$$

$$\left| \frac{(-1)^{n+1}}{(2n)!} \cdot \frac{1}{3^{2n-1}(2n-1)} \right| < 10^{-3} \Rightarrow \frac{1}{(2n)!} \cdot \frac{1}{3^{2n-1}(2n-1)} < \frac{1}{1000} \Rightarrow$$

$$\Rightarrow 1000 < (2n)!\, 3^{2n-1}(2n-1)$$

$n=2 \Rightarrow 4! \cdot 3^3 \cdot 3 = 24 \cdot 9 \cdot 9 = 24 \cdot 81 = 1944 > 1000$ adev.

$n=1 \Rightarrow 2! \cdot 3 \cdot 1 = 6 < 1000$ fals

$$S \approx S_1 = \frac{1}{2} \cdot \frac{1}{3 \cdot 1} = \frac{1}{6}$$

4) Calculati polinoamele Taylor de ordinul I și II centrate în $(1,-1)$ pt. funcția $f(x,y) = y^2 e^{x-y^2}$

$$\frac{\partial f}{\partial x} = y^2 e^{x-y^2}$$

$$\frac{\partial f}{\partial y} = 2y e^{x-y^2} + y^2 \cdot e^{x-y^2}(-2y) = 2y e^{x-y^2}[1-y^2]$$

$$T_1(1,-1) = f(1,-1) + \frac{1}{1!}\left[\frac{\partial f}{\partial x}(1,-1)(x-1) + \frac{\partial f}{\partial y}(1,-1)(y+1)\right]$$

$$= (-1)^2 \cdot e^{1-(-1)^2} + 1^2 e^{1-(-1)^2}(x-1) + 2\cdot(-1)\cdot e^{1-(-1)^2}[1-(-1)^2](y+1)$$

$$= 1 \cdot 1 + 1 \cdot 1 \cdot (x-1) + (-2) \cdot 1 \cdot 0 \,(y+1)$$

$$= 1 + (x-1) = \cancel{1} + x - \cancel{1} = x$$

$$\frac{\partial^2 f}{\partial x^2} = y^2 e^{x-y^2}$$

$$\frac{\partial^2 f}{\partial y^2} = 2e^{x-y^2}(1-y^2) + 2y e^{x-y^2}(-2y)(1-y^2) + 2y e^{x-y^2}(-2y)$$

$$= 2e^{x-y^2}(1-y^2-2y^2+2y^4-2y^2) = 2e^{x-y^2}(1-5y^2+2y^4)$$

$$\frac{\partial^2 f}{\partial y \partial x} = 2y e^{x-y^2}(1-y^2)$$

$$T_2(1;-1) = T_1 + \frac{1}{2!}\left[\frac{\partial^2 f}{\partial x^2}(1;-1)(x-1)^2 + 2\frac{\partial^2 f}{\partial x \partial y}(1;-1)(x-1)(y+1)\right.$$

$$\left. + \frac{\partial^2 f}{\partial y^2}(1;-1)(y+1)^2\right] = x + \frac{1}{2!}\{(-1)^2 \cdot e^{1-(-1)^2}(x-1)^2 +$$

$$+2\cdot 2(-1)\cdot e^{1-(-1)^2}[1-(-1)^2](x-1)(y-1) + 2\cdot e^{1-(-1)^2}(1-5\cdot(-1)^2 +$$

$$+2\cdot(-1)^4](y+1)^2\} = x + \frac{1}{2}\left[x^2-2x-1+2\cdot(-2)(y^2+2y+1)\right]$$

$$= \cancel{x} + \frac{x^2}{2} - \cancel{x} - \frac{1}{2} - 2(y^2+2y+1)$$

$$= -2y-2 + \frac{x^2}{2} - \frac{1}{2} - 2y^2 - 4y - 2 = \frac{x^2}{2} - 2y^2 - 6y - \frac{9}{2}$$

5) $X = \{x_1 x_2 x_3 \ldots \mid x_i \in \{a,b,c\}\}$

$Y = \left\{\frac{\sqrt{x^2-2}}{x^2+\sqrt{2}}, x \in \mathbb{N}, x \geq 2\right\}$ Demonstrați că X nu este numărabilă, dar Y este.

Y = o mulțime în care putem asocia o f.

$$f: \mathbb{N}_{\geq 2} \longrightarrow \mathbb{R}, \quad f(x) = \frac{\sqrt{x^2-2}}{x^2+\sqrt{2}}$$

Deci prin numirea unei valori $x \in \mathbb{N}, x \geq 2$ avem o unică valoare $f(x)$ corespunzătoare

$\Rightarrow Y$ este echipotentă cu $\mathbb{N}$ $\Rightarrow Y$ este mulțime numărabilă

$\mathbb{N}$ mulțime numărabilă

Presupunem că X este o mulțime numărabilă

fie $q_1 = x_1^1 x_1^2 x_1^3 \ldots$ elemente ale mulțimii X

$q_2 = x_2^1 x_2^2 x_2^3 \ldots$ adică $x_i^i \in \{a, b, c\}$

$\vdots$

fie $r = y_1 y_2 y_3 \ldots$

a.î. $y_1 \neq x_1^1$

$y_2 \neq x_2^2$

$\vdots$

$y_i \in \{a, b, c\}$

$\Rightarrow r \neq q_1$

$r \neq q_2$

$\vdots$

$\Rightarrow r \notin X \Rightarrow X$ nu este o mulțime numărabilă

Partial drive 312 CC - outline

① Natura seriei $\sum_{n\geq 1} \frac{2\cdot 5\cdot 8\cdot \ldots\cdot (3n-1)}{1\cdot 5\cdot 9\cdot \ldots\cdot (4n-3)} \cdot \frac{1}{2^n(3n+1)}$

$$\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = \lim_{n\to\infty} \frac{2\cdot 5\cdot \ldots\cdot (3n-1)(3n+2)}{1\cdot 5\cdot \ldots\cdot (4n-3)(4n+1)} \cdot \frac{1\cdot 5\cdot \ldots\cdot (4n-3)}{2\cdot 5\cdot \ldots\cdot (3n-1)} \cdot \frac{2^n(3n+1)}{2^{n+1}(3n+4)}$$

$$= \lim_{n\to\infty} \frac{3n+2}{4n+1}\cdot\frac{3n+1}{3n+4}\cdot\frac{1}{2} = \frac{1}{2}\cdot\frac{3}{4}\cdot\frac{3}{3} = \frac{3}{8} < 1$$

$$\xRightarrow[\text{Rap.}]{\text{Crit.}} \sum_{n\geq 1} \frac{2\cdot 5\cdot \ldots\cdot (3n-1)}{1\cdot 5\cdot \ldots\cdot (4n-3)} \cdot \frac{1}{2^n(3n+1)} \quad \text{converg.}$$

② Natura seriei în funcție de $a \in (0, 19)$

$$\sum_{n\geq 1} (-1)^n \frac{\sqrt{n^{19}+20n^a+18} - n^9\sqrt{n}}{n^2} = \sum_{n\geq 1} x_n$$

$$x_n = (-1)^n \frac{\sqrt{n^{19}+20n^a+18} - n^9\sqrt{n}}{n^2} = (-1)^n \frac{n^{19}+20n^a+18-n^{19}}{n^2\left(\sqrt{n^{19}+20n^a+18} + n^9\sqrt{n}\right)}$$

$$x_n = (-1)^n \frac{n^a\left(20+\frac{18}{n^a}\right)}{n^{11}\sqrt{n}\left(\sqrt{1+\frac{20}{n^{a-19}}+\frac{18}{n^{19}}}+1\right)}$$

fie $y_n = \frac{1}{n^{11+1/2-a}}$

$$\lim_{n\to\infty} \frac{|x_n|}{y_n} = \lim_{n\to\infty} \frac{n^a\left(20+\frac{18}{n^a}\right)}{n^{11}\sqrt{n}\left(\sqrt{1+\frac{20}{n^{a-19}}+\frac{18}{n^{19}}}+1\right)} \cdot n^{11+1/2-a} = \frac{20}{2}$$

$$\Rightarrow \lim_{n\to\infty} \frac{|x_n|}{y_n} = 10 \in (0, +\infty) \xRightarrow[\text{comp la lim. } n\geq 1]{\text{Crit. de}} \sum |x_n| \sim \sum_{n\geq 1} y_n$$

Dar y_n serie Riemann $\neq 1$

I pt $11+1/2-a>1 \Rightarrow 10+1/2>a \Rightarrow a<\frac{21}{2}$

$\Rightarrow$ pt. $a\in(0,\frac{21}{2}) \Rightarrow \sum_{n\geq 1} x_n$ absolut converg $\Rightarrow \sum_{n\geq 1} x_n$ converg

II pt. $11+1/2-a\leq 1 \Rightarrow a\geq \frac{21}{2} \Rightarrow$

$\Rightarrow$ pt. $a\in\left[\frac{21}{2};19\right) \Rightarrow \sum_{n\geq 1}|x_n|$ div. $\Rightarrow \sum_{n\geq 1} x_n$ nu e absolut converg.

$$\alpha_n=\frac{n^a\left(20+\frac{18}{n^a}\right)}{n^{11}\sqrt{n}\left(\sqrt{1+\frac{20}{n^{a-19}}+\frac{18}{n^{19}}}+1\right)}$$

$$\lim_{n\to\infty}\alpha_n=\begin{cases}0 & \text{pt. } a\in\left[\frac{21}{2};\frac{23}{2}\right)\\ 10 & \text{pt. } a=\frac{23}{2}\\ \infty & \text{pt. } a\in\left(\frac{23}{2};19\right)\end{cases}\xrightarrow{\text{Crit. nec.}}\sum_{n\geq 1} x_n \text{ divergentă pt. } a\in\left[\frac{23}{2};19\right)$$

pt $a\in\left[\frac{21}{2};\frac{23}{2}\right)$ $\alpha_n\searrow$

$\lim_{n\to\infty}\alpha_n=0$

$\alpha_n>0$

$\xrightarrow{\text{Crit. Leibniz}}\sum_{n\geq 1} x_n$ converg. pt. $a\in\left[\frac{21}{2};\frac{23}{2}\right]$

3) Aflați mulțimea de convergență și suma seriei

$$\sum_{n\geq 2}\frac{(-1)^n}{n}\cdot x^{2n},\ x\in\mathbb{R}$$

$$a_n=\frac{(-1)^n}{n}$$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\left|\frac{(-1)^{n+1}}{n+1}\cdot\frac{n}{(-1)^n}\right|=\lim_{n\to\infty}\left|\frac{-n}{n+1}\right|=1\Rightarrow$$

$\Rightarrow$ $\boxed{I_c = (-1,1)}$

$x = \pm 1 \Rightarrow \sum_{n\geq 2} \frac{(-1)^n}{n} \cdot (\pm 1)^{2n} = \sum_{n\geq 2} \frac{(-1)^n}{n}$

fie $\alpha_n = \frac{1}{n}$ $\quad \alpha_n \geq 0$ $\quad \alpha_n \searrow$ $\quad \lim_{n\to\infty} \alpha_n = 0$ $\Rightarrow$ Crit. Leibniz $\Rightarrow \sum_{n\geq 2} (-1)^n \cdot \frac{1}{n}$ converg

$\Rightarrow \boxed{D_c = [-1;1]}$

Notăm $x^2 = y$

$S(x) = \sum_{n\geq 2} \frac{(-1)^n}{n} x^{2n} \Rightarrow f(y) = \sum_{n\geq 2} \frac{(-1)^n}{n} y^n \xrightarrow{\text{deriv.}}$

$\xrightarrow{\text{deriv.}} f'(y) = \sum_{n\geq 2} \frac{(-1)^n}{n} \cdot n y^{n-1} = \sum_{n\geq 2} (-1)^n \cdot y^{n-1} = \sum_{n\geq 1} (-1)^{n+1} y^n$

Știm $\sum_{n\geq 0} (-1)^n y^n = \frac{1}{1+y} \Rightarrow \sum_{n\geq 0} (-1)^{n+1} y^n = \frac{-1}{1+y} \Rightarrow$

$\Rightarrow -1 + \sum_{n\geq 1} (-1)^{n+1} y^n = \frac{-1}{1+y} \Rightarrow \sum_{n\geq 1} (-1)^{n+1} y^n = 1 - \frac{1}{1+y}$

$\Rightarrow f'(y) = 1 - \frac{1}{1+y} \xrightarrow{\text{integ.}} f(y) = y - \ln(1+y)$

$\boxed{S(x) = x^2 - \ln(1+x^2)}$

(4) Calculați $Df(1,1,0)$ pt. funcția

$f: \mathbb{R}^3 \longrightarrow \mathbb{R}^2 \quad f(x,y,z) = \left(\underbrace{\sqrt{4y^2+z^2+1}}_{f_1}, \underbrace{\ln(x^2+3z^4+2)}_{f_2}\right)$

$$\frac{\partial f_1}{\partial x} = 0$$

$$\frac{\partial f_1}{\partial y} = \frac{8y}{2\sqrt{4y^2+z^2+1}} = \frac{4y}{\sqrt{4y^2+z^2+1}} = \frac{4}{\sqrt{5}}$$

$$\frac{\partial f_1}{\partial z} = \frac{2z}{2\sqrt{4y^2+z^2+1}} = \frac{z}{\sqrt{4y^2+z^2+1}} = 0$$

$$\frac{\partial f_2}{\partial x} = \frac{2x}{x^2+3z^4+2} = \frac{2}{3}$$

$$\frac{\partial f_2}{\partial y} = 0$$

$$\frac{\partial f_2}{\partial z} = \frac{12z^3}{x^2+3z^4+2} = 0$$

$$\mathcal{M}_f(1,1,0) = \begin{pmatrix} 0 & \frac{4}{\sqrt{5}} & 0 \\ \frac{2}{3} & 0 & 0 \end{pmatrix}$$

$$Df(1,1,0) = \begin{pmatrix} 0 & \frac{4}{\sqrt{5}} & 0 \\ \frac{2}{3} & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \frac{4y}{\sqrt{5}} \\ \frac{2x}{3} \end{pmatrix}$$

Partial drive – 2022

① Studiați natura seriei

a) $\sum_{n\geq 1} \frac{a(a+1)(a+2)\dots(a+n-1)}{n^2 \cdot n!}$ $\quad a>0$

$$\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = \lim_{n\to\infty} \frac{a(a+1)\dots(a+n-1)(a+n)}{(n+1)^2(n+1)!} \cdot \frac{n^2 \cdot n!}{a(a+1)\dots(a+n-1)}$$

$$= \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^2 \cdot \frac{a+n}{n+1} = 1 \xrightarrow[\text{Rap.}]{\text{Crit.}} \text{nu se poate determ.}$$

$$\lim_{n\to\infty} n\left[\frac{x_n}{x_{n+1}} - 1\right] = \lim_{n\to\infty} n\left[\left(\frac{n+1}{n}\right)^2 \cdot \frac{n+1}{a+n} - 1\right] =$$

$$= \lim_{n\to\infty} n \frac{(n+1)^3 - n^2(a+n)}{n^2(a+n)} = \lim_{n\to\infty} \frac{n^3+3n^2+3n+1-an^2-n^3}{n(a+n)}$$

$$= \lim_{n\to\infty} \frac{n^2(3-a)+3n+1}{n^2+an} = \frac{3-a}{1} = 3-a$$

$\xrightarrow[\text{R-D}]{\text{Crit.}}$ pt. $3-a<1 \Rightarrow a>2 \Rightarrow \sum_{n\geq 1} x_n$ diverg.

pt. $3-a>1 \Rightarrow a<2 \Rightarrow \sum_{n\geq 1} x_n$ converg.

pt. $3-a=1 \Rightarrow a=2 \Rightarrow \sum_{n\geq 1} x_n = \sum_{n\geq 1} \frac{2\cdot 3\cdot \dots \cdot (n+1)}{n^2 \cdot n!} =$

$$= \sum_{n\geq 1} \frac{(n+1)!}{n^2 \cdot n} = \sum_{n\geq 1} \frac{n+1}{n^2}$$

fie $y_n = \frac{1}{n} \Rightarrow \sum_{n\geq 1} y_n$ serie Riemann diverg.

$$\lim_{n\to\infty} \frac{x_n}{y_n} = \lim_{n\to\infty} \frac{n+1}{n^2} \cdot n = 1 \in (0;+\infty) \xrightarrow[\text{la lim.}]{\text{Crit. de comp}}$$

$$\Rightarrow \sum_{n\geq 1} x_n \sim \sum_{n\geq 1} y_n \Rightarrow \sum_{n\geq 1} x_n \text{ diverg. pt. } a=2$$

b) $\sum_{n\geq 1} \frac{(-1)^n}{n(\sqrt{n^2+n}-n)}$

fie $\alpha_n = \frac{1}{n(\sqrt{n^2+n}-n)} = \frac{\sqrt{n^2+n}+n}{n(n^2+n-n^2)} = \frac{\sqrt{n^2+n}+n}{n^2}$

$\lim_{n\to\infty} \alpha_n = \lim_{n\to\infty} \frac{x(\sqrt{1+\frac{1}{n}}+1)}{n^2} = \frac{2}{\infty} = 0$

$\alpha_n > 0$ pt. că $n \geq 1$, iar α_n este o serie de nr. poz.

$\alpha_1 = \frac{\sqrt{2}+1}{1} = 1+\sqrt{2} \simeq 2{,}41$ $\qquad \alpha_2 = \frac{\sqrt{6}+2}{4} \simeq 1{,}11$

$P(k): \alpha_k = \frac{\sqrt{k(k+1)}+k}{k^2} = \sqrt{\frac{k+1}{k^3}} + \frac{1}{k}$

$P(k+1)\ \alpha_{k+1} = \frac{\sqrt{(k+1)(k+2)}+k+1}{(k+1)^2} = \sqrt{\frac{k+2}{(k+1)^3}} + \frac{1}{k+1}$

$\frac{1}{k} > \frac{1}{k+1}$

$\frac{(k+1)^3}{k^3} > \frac{k+1}{k} = 1+\frac{1}{k} > 1+\frac{1}{k+1} = \frac{k+2}{k+1} \Rightarrow \frac{k+1}{k^3} > \frac{k+2}{(k+1)^3}$

$\Rightarrow \sqrt{\frac{k+1}{k^3}} + \frac{1}{k} > \sqrt{\frac{k+2}{(k+1)^3}} + \frac{1}{k+1} \Rightarrow \alpha_k > \alpha_{k+1} \xrightarrow{\text{IMC}}$

$\Rightarrow \alpha_n > \alpha_{n+1} \Rightarrow \frac{\alpha_{n+1}}{\alpha_n} < 1 \Rightarrow \alpha_n \downarrow$

$\alpha_n > 0$

$\lim_{n\to\infty} \alpha_n = 0$ $\xrightarrow[\text{Leibniz}]{\text{Crit.}}$ $\sum_{n\geq 1} (-1)^n \cdot \alpha_n$ convergent $\Rightarrow$

$\alpha_n \downarrow$

$\Rightarrow \sum_{n\geq 1} x_n$ conv.

② Să se determine mulțimea de convergență și suma seriei $\sum_{n\geq 1}(-1)^n \cdot \frac{x^{2n+1}}{2n+1}$

$$\ell = \lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{(-1)^{n+1}}{2n+3}\cdot\frac{2n+1}{(-1)^n}\right| = 1 \Rightarrow R = \frac{1}{\ell} = 1$$

$$\Rightarrow I_c = (-1;1)$$

$$x = 1 \Rightarrow \sum_{n\geq 1} x_n = \sum_{n\geq 1}(-1)^n \cdot \frac{1}{2n+1} = \sum_{n\geq 1}\frac{(-1)^n}{2n+1}$$

fie $\alpha_n = \frac{1}{2n+1} \Rightarrow \alpha_n > 0$

$\lim_{n\to\infty} \alpha_n = 0$

$\frac{\alpha_{n+1}}{\alpha_n} = \frac{2n+1}{2n+3} < 1 \Rightarrow \alpha_n \downarrow$

$\xrightarrow{\text{Crit. Leibniz}} \sum_{n\geq 1}\frac{(-1)^n}{(2n+1)}$ conv.

$$x = -1 \Rightarrow \sum_{n\geq 1} x_n = \sum_{n\geq 1}(-1)^n \cdot \frac{-1}{2n+1} = -\sum_{n\geq 1}(-1)^n \cdot \frac{1}{2n+1} \Longrightarrow$$

Conform celor de mai sus $\Rightarrow \sum_{n\geq 1}(-1)^n \cdot \frac{1}{2n+1}$ converg.

$\Rightarrow \sum_{n\geq 1} x_n$ converg.

$\Longrightarrow \boxed{M_c = [-1;1]}$

$$S(x) = \sum_{n\geq 1}(-1)^n \cdot \frac{x^{2n+1}}{2n+1} \xrightarrow{\text{deriv.}} S'(x) = \sum_{n\geq 1}(-1)^n \cancel{(2n+1)} \cdot \frac{x^{2n}}{\cancel{2n+1}}$$

$$\Rightarrow S'(x) = \sum_{n\geq 1}(-1)^n \cdot x^{2n}$$

Notăm $x^2 = y$ $\Rightarrow f'(y) = \sum_{n\geq 1}(-1)^n y^n \Longrightarrow$

$$\sum_{n\geq 0}(-1)^n y^n = \frac{1}{1+y} \Rightarrow 1 + \sum_{n\geq 1}(-1)^n y^n = \frac{1}{1+y}$$

$$\Rightarrow f'(y)=\frac{1}{1+y}-1 \Rightarrow S'(x)=\frac{1}{1+x^2}-1 \xrightarrow{\text{integ.}}$$

$$\xrightarrow{\text{integ.}} \boxed{S(x)=\operatorname{arctg} x - x}$$

(3) Să se aproximeze cu o eroare mai mică decât 10^{-2} valoarea integralei $\int_0^1 \frac{1-e^{-x^2}}{x^2}\,dx$

$$e^x=\sum_{n\geq 0}\frac{1}{n!}\cdot x^n \Rightarrow e^{-x^2}=\sum_{n\geq 0}\frac{1}{n!}(-x^2)^n$$

$$\frac{1-e^{-x^2}}{x^2}=\frac{1-\sum_{n\geq 0}\frac{1}{n!}(-x^2)^n}{x^2}=\frac{\cancel{1}-\cancel{1}+\sum_{n\geq 1}\frac{1}{n!}(-1)^n x^{2n}}{x^2}=$$

$$=\sum_{n\geq 1}\frac{1}{n!}(-1)^n\cdot x^{2n-2}$$

$$\int_0^1 \frac{1-e^{-x^2}}{x^2}\,dx=\sum_{n\geq 1}\frac{1}{n!}(-1)^n\int_0^1 x^{2n-2}\,dx=\sum_{n\geq 1}\frac{1}{n!}(-1)^n\cdot\frac{x^{2n-1}}{2n-1}\Big|_0^1$$

$$=\sum_{n\geq 1}\frac{1}{n!}(-1)^n\cdot\frac{1}{2n-1}$$

$$\left|\frac{(-1)^n}{n!}\cdot\frac{1}{2n-1}\right|<10^{-2} \Rightarrow 100<n!(2n-1)$$

pt. $n=4 \Rightarrow 100<4!\cdot 7 \Rightarrow 100<24\cdot 7 \Rightarrow 100<168$ ✓

pt. $n=3 \Rightarrow 100<3!\cdot 5 \Rightarrow 100<30$ fals ⇒

$$S\approx S_3=\frac{1}{1!}(-1)\cdot\frac{1}{1}+\frac{1}{2!}(-1)^2\cdot\frac{1}{3}+\frac{1}{3!}(-1)^3\cdot\frac{1}{5}$$

$$=-1+\frac{1}{6}-\frac{1}{30}=\frac{-30+5-1}{30}=\frac{-26}{30}=\frac{-13}{15}\approx -0{,}87$$

(4) Să se determine punctele de extrem local ale funcției

$$f(x,y,z)=\frac{1}{z}+\frac{z}{x}+\frac{x}{y}+y-1 \qquad x,y,z\neq 0$$

$$\frac{\partial f}{\partial x}=2\cdot\frac{(-1)}{x^2}+\frac{1}{y}=\frac{-z}{x^2}+\frac{1}{y}$$

$$\frac{\partial f}{\partial y}=x\cdot\frac{(-1)}{y^2}+1=\frac{-x}{y^2}+1$$

$$\frac{\partial f}{\partial z}=-\frac{1}{z^2}+\frac{1}{x}$$

$$\begin{cases}\frac{-z}{x^2}+\frac{1}{y}=0\\ \frac{-x}{y^2}+1=0\\ -\frac{1}{z^2}+\frac{1}{x}=0\end{cases}\Rightarrow\begin{cases}\frac{z}{x^2}=\frac{1}{y}\\ x=y^2\\ x=z^2\end{cases}$$

$$\Rightarrow\begin{cases}zy=x^2\Rightarrow x=x^2\Rightarrow x(x-1)=0\\ y=\sqrt{x}\\ z=\sqrt{x}\end{cases}\qquad x,y,z\neq 0$$

$$\Rightarrow x=1\Rightarrow y=1,\ z=1$$

$$\Rightarrow \underline{M_1(1;1;1)}$$

$$\frac{\partial^2 f}{\partial x^2}=(-z)(-2)\frac{1}{x^3}=\frac{2z}{x^3}=2$$

$$\frac{\partial^2 f}{\partial y^2}=(-x)(-2)\frac{1}{y^3}=\frac{2x}{y^3}=2$$

$$\frac{\partial^2 f}{\partial z^2}=(-1)\cdot(-2)\cdot\frac{1}{z^3}=\frac{2}{z^3}=2$$

$$\frac{\partial^2 f}{\partial x\partial y}=-\frac{1}{y^2}=-1$$

$$\frac{\partial^2 f}{\partial x\partial z}=-\frac{1}{x^2}=-1$$

$$\frac{\partial^2 f}{\partial z\partial y}=0=0$$

$$H_f(1;1;1)=\begin{pmatrix}2&-1&-1\\-1&2&0\\-1&0&2\end{pmatrix}$$

$$\Delta_1=2>0$$

$$\Delta_2=4-1=3>0$$

$$\Delta_3=\begin{vmatrix}2&-1&-1\\-1&2&0\\-1&0&2\end{vmatrix}=8-2-2=4>0$$

$$\Rightarrow$$

$$\Rightarrow \underline{(1;1;1)\text{ pct. de minim local}}$$

(5) Să se determine polinomul Taylor de ordinul al II-lea pt. funcția $f(x,y)=\sqrt{x^2+y^2}$ centrat în $(1,0)$

$$\frac{\partial f}{\partial x}=\frac{2x}{2\sqrt{x^2+y^2}}=\frac{x}{\sqrt{x^2+y^2}} \Rightarrow \frac{\partial f}{\partial x}(1,0)=1$$

$$\frac{\partial f}{\partial y}=\frac{2y}{2\sqrt{x^2+y^2}}=\frac{y}{\sqrt{x^2+y^2}} \Rightarrow \frac{\partial f}{\partial y}(1,0)=0$$

$$\frac{\partial^2 f}{\partial x^2}=\frac{\sqrt{x^2+y^2}-x\cdot\frac{2x}{2\sqrt{x^2+y^2}}}{x^2+y^2}=\frac{x^2+y^2-x^2}{(x^2+y^2)\sqrt{x^2+y^2}}=\frac{y^2}{(x^2+y^2)\sqrt{x^2+y^2}}$$

$$\frac{\partial^2 f}{\partial x^2}(1,0)=0$$

$$\frac{\partial^2 f}{\partial y^2}=\frac{\sqrt{x^2+y^2}-y\cdot\frac{2y}{2\sqrt{x^2+y^2}}}{x^2+y^2}=\frac{x^2+y^2-y^2}{(x^2+y^2)\sqrt{x^2+y^2}}=\frac{x^2}{(x^2+y^2)\sqrt{x^2+y^2}}$$

$$\frac{\partial^2 f}{\partial y^2}(1,0)=\frac{1}{1\cdot\sqrt{1}}=1$$

$$\frac{\partial^2 f}{\partial x\,\partial y}=\frac{\partial}{\partial x}\left(\frac{y}{\sqrt{x^2+y^2}}\right)=y\cdot\left(-\frac{1}{2}\right)\cdot(x^2+y^2)^{-3/2}\cdot 2x=\frac{-xy}{(x^2+y^2)^{3/2}}$$

$$\frac{\partial^2 f}{\partial x\,\partial y}(1,0)=0$$

$$f(1,0)=\sqrt{1^2+0^2}=1$$

$$T_2(1,0) = f(1,0) + \frac{1}{1!}\left[\frac{\partial f}{\partial x}(1,0)(x-1) + \frac{\partial f}{\partial y}(1,0)(y-0)\right] +$$

$$+ \frac{1}{2!}\left[\frac{\partial^2 f}{\partial x^2}(1,0)(x-1)^2 + \frac{\partial^2 f}{\partial y^2}(1,0)(y-0)^2 + 2\frac{\partial^2 f}{\partial x \partial y}(1,0)(x-1)(y-0)\right]$$

$$T_2(1,0) = 1 + 1(x-1) + 0 \cdot y + \frac{1}{2}\left[0(x-1)^2 + 1 \cdot y^2 + 2 \cdot 0 (x-1)y\right]$$

$$T_2(1,0) = 1 + x - 1 + \frac{1}{2}y^2 = x + \frac{1}{2}y^2$$